EXP 24

1. Create a Library Management System GUI connected with database

SQL Code:

```
USE SIT_CSE;

-- Table for Q1: Library Management System
CREATE TABLE IF NOT EXISTS books (
    book_id INT PRIMARY KEY AUTO_INCREMENT,
    title VARCHAR(100) NOT NULL,
    author VARCHAR(100) NOT NULL,
    publisher VARCHAR(100) NOT NULL,
    price DECIMAL(10, 2) NOT NULL
);


-- Sample Data
INSERT INTO books (title, author, publisher, price) VALUES
('Java Programming', 'James Gosling', 'Oracle Press', 550.00),
('Data Structures', 'Mark Allen Weiss', 'Pearson', 620.00);
```

Java Code:

```
import javax.swing.*;
import javax.swing.table.DefaultTableModel;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.sql.*;

public class Q1_LibraryManagementGUI extends JFrame {
    private JTextField txtTitle, txtAuthor, txtPublisher, txtPrice;
    private JTable table;
    private DefaultTableModel model;

    private static final String URL = "jdbc:mysql://localhost:3306/SIT_CSE";
```

```
    private static final String USER = "root";
    private static final String PASSWORD = "root";

    public Q1_LibraryManagementGUI() {
        setTitle("Library Management System");
        setSize(700, 500);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);
        setLayout(new BorderLayout(10, 10));

        // Input Panel
        JPanel inputPanel = new JPanel(new GridLayout(5, 2, 5, 5));
        inputPanel.setBorder(BorderFactory.createTitledBorder("Book Details"));

        inputPanel.add(new JLabel("Book Title:"));
        txtTitle = new JTextField();
        inputPanel.add(txtTitle);

        inputPanel.add(new JLabel("Author:"));
        txtAuthor = new JTextField();
        inputPanel.add(txtAuthor);

        inputPanel.add(new JLabel("Publisher:"));
        txtPublisher = new JTextField();
        inputPanel.add(txtPublisher);

        inputPanel.add(new JLabel("Price:"));
        txtPrice = new JTextField();
        inputPanel.add(txtPrice);

        JButton btnAdd = new JButton("Add Book");
        JButton btnFetch = new JButton("View Books");
        inputPanel.add(btnAdd);
        inputPanel.add(btnFetch);

        add(inputPanel, BorderLayout.NORTH);

        // Table Panel
        model = new DefaultTableModel(new String[]{"ID", "Title", "Author", "Publisher",
"Price"}, 0);
        table = new JTable(model);
        add(new JScrollPane(table), BorderLayout.CENTER);

        // Event Listeners
```

```
        btnAdd.addActionListener((ActionEvent e) -> addBook());
        btnFetch.addActionListener((ActionEvent e) -> loadBooks());

        loadBooks();
    }

    private void addBook() {
        String query = "INSERT INTO books (title, author, publisher, price) VALUES (?,
?, ?, ?)";
        try (Connection con = DriverManager.getConnection(URL, USER,
PASSWORD);
             PreparedStatement ps = con.prepareStatement(query)) {

            ps.setString(1, txtTitle.getText());
            ps.setString(2, txtAuthor.getText());
            ps.setString(3, txtPublisher.getText());
            ps.setDouble(4, Double.parseDouble(txtPrice.getText()));

            ps.executeUpdate();
            JOptionPane.showMessageDialog(this, "Book Added Successfully!");
            txtTitle.setText(""); txtAuthor.setText(""); txtPublisher.setText("");
txtPrice.setText("");
            loadBooks();
        } catch (Exception ex) {
            JOptionPane.showMessageDialog(this, "Error: " + ex.getMessage());
        }
    }

    private void loadBooks() {
        model.setRowCount(0);
        String query = "SELECT * FROM books";
        try (Connection con = DriverManager.getConnection(URL, USER,
PASSWORD);
             Statement stmt = con.createStatement();
             ResultSet rs = stmt.executeQuery(query)) {

            while (rs.next()) {
                model.addRow(new Object[]{
                        rs.getInt("book_id"),
                        rs.getString("title"),
                        rs.getString("author"),
                        rs.getString("publisher"),
                        rs.getDouble("price")
                });
```

```
            }
        } catch (Exception ex) {
            JOptionPane.showMessageDialog(this, "Error loading records: " +
ex.getMessage());
        }
    }

    public static void main(String[] args) {
        SwingUtilities.invokeLater(() -> new
Q1_LibraryManagementGUI().setVisible(true));
    }
}
```

2. Create a Book Issue Tracking System GUI application using Java Swing connected to a database. The system should store book ID, student name, issue date, and return date, and allow record management.

SQL Code:

```
-- Table for Q2: Book Issue Tracking System
CREATE TABLE IF NOT EXISTS book_issues (
    issue_id INT PRIMARY KEY AUTO_INCREMENT,
    book_id INT NOT NULL,
    student_name VARCHAR(100) NOT NULL,
    issue_date DATE NOT NULL,
    return_date DATE NOT NULL
);

-- Sample Data
INSERT INTO book_issues (book_id, student_name, issue_date, return_date)
VALUES
(101, 'Akshat Sharma', '2026-03-01', '2026-03-15'),
(102, 'Rohan Verma', '2026-03-05', '2026-03-19');
```

Java Code:

```
import javax.swing.*;
import javax.swing.table.DefaultTableModel;
import java.awt.*;
```

```
import java.awt.event.ActionEvent;
import java.sql.*;

public class Q2_BookIssueTrackerGUI extends JFrame {
    private JTextField txtBookID, txtStudentName, txtIssueDate, txtReturnDate;
    private JTable table;
    private DefaultTableModel model;

    private static final String URL = "jdbc:mysql://localhost:3306/SIT_CSE";
    private static final String USER = "root";
    private static final String PASSWORD = "root";

    public Q2_BookIssueTrackerGUI() {
        setTitle("Book Issue Tracking System");
        setSize(750, 500);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);
        setLayout(new BorderLayout(10, 10));

        // Form Panel
        JPanel formPanel = new JPanel(new GridLayout(5, 2, 5, 5));
        formPanel.setBorder(BorderFactory.createTitledBorder("Issue Details"));

        formPanel.add(new JLabel("Book ID:"));
        txtBookID = new JTextField();
        formPanel.add(txtBookID);

        formPanel.add(new JLabel("Student Name:"));
        txtStudentName = new JTextField();
        formPanel.add(txtStudentName);

        formPanel.add(new JLabel("Issue Date (YYYY-MM-DD):"));
        txtIssueDate = new JTextField();
        formPanel.add(txtIssueDate);

        formPanel.add(new JLabel("Return Date (YYYY-MM-DD):"));
        txtReturnDate = new JTextField();
        formPanel.add(txtReturnDate);

        JButton btnIssue = new JButton("Issue Book");
        JButton btnRefresh = new JButton("Refresh Table");
        formPanel.add(btnIssue);
        formPanel.add(btnRefresh);
```

```
        add(formPanel, BorderLayout.NORTH);

        // Table Panel
        model = new DefaultTableModel(new String[]{"Issue ID", "Book ID", "Student
Name", "Issue Date", "Return Date"}, 0);
        table = new JTable(model);
        add(new JScrollPane(table), BorderLayout.CENTER);

        // Actions
        btnIssue.addActionListener((ActionEvent e) -> issueBook());
        btnRefresh.addActionListener((ActionEvent e) -> loadIssueRecords());

        loadIssueRecords();
    }

    private void issueBook() {
        String query = "INSERT INTO book_issues (book_id, student_name,
issue_date, return_date) VALUES (?, ?, ?, ?)";
        try (Connection con = DriverManager.getConnection(URL, USER,
PASSWORD);
             PreparedStatement ps = con.prepareStatement(query)) {

            ps.setInt(1, Integer.parseInt(txtBookID.getText()));
            ps.setString(2, txtStudentName.getText());
            ps.setDate(3, Date.valueOf(txtIssueDate.getText()));
            ps.setDate(4, Date.valueOf(txtReturnDate.getText()));

            ps.executeUpdate();
            JOptionPane.showMessageDialog(this, "Book Issued Successfully!");
            txtBookID.setText(""); txtStudentName.setText(""); txtIssueDate.setText("");
txtReturnDate.setText("");
            loadIssueRecords();
        } catch (Exception ex) {
            JOptionPane.showMessageDialog(this, "Error: " + ex.getMessage());
        }
    }

    private void loadIssueRecords() {
        model.setRowCount(0);
        String query = "SELECT * FROM book_issues";
        try (Connection con = DriverManager.getConnection(URL, USER,
PASSWORD);
             Statement stmt = con.createStatement();
             ResultSet rs = stmt.executeQuery(query)) {
```

```
            while (rs.next()) {
                model.addRow(new Object[]{
                        rs.getInt("issue_id"),
                        rs.getInt("book_id"),
                        rs.getString("student_name"),
                        rs.getDate("issue_date"),
                        rs.getDate("return_date")
                });
            }
        } catch (Exception ex) {
            JOptionPane.showMessageDialog(this, "Error loading records: " +
ex.getMessage());
        }
    }

    public static void main(String[] args) {
        SwingUtilities.invokeLater(() -> new
Q2_BookIssueTrackerGUI().setVisible(true));
    }
}
```